# BHASKARACHARYA NATIONAL INSTITUTE FOR SPACE APPLICATIONS AND GEO-INFORMATICS

**WEEKLY PROGRESS REPORT (05/10/2026 – 11/10/2026)**

**WEEK 11**

| PROJECT NAME | CHATBOT FOR STRUCTURED, UNSTRUCTURED AND SEMANTIC KNOWLEDGE QUERY |
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| **PROJECT DESCRIPTION :** | DESIGN AND IMPLEMENT A CHATBOT THAT CAN ANSWER TO DIFFERENT TYPE OF QUERY LIKE STRUCTURED, UNSTRUCTURED AND SEMANTIC KNOWLEDGE BY DECIDING OWN WHAT TYPE OF QUERY BY |
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| **05/10/2026** | • Reviewed the existing architecture and planned a unified organization knowledge index for documents, images, and structured data.<br>• Identified the general-knowledge response bypass and defined requirements for answering only from organizational evidence. |
| **06/10/2026** | • Implemented unified indexing of document content, image descriptions/OCR, and table context with organization and source metadata.<br>• Replaced the default query-classifier flow with organization-scoped retrieval. |
| **07/10/2026** | • Added evidence checks and refusal handling for questions unsupported by organizational data.<br>• Implemented prompt-injection filtering and answer-validation checks to improve grounded responses. |
| **08/09/2026** | • Retained SQL execution for exact calculations and added table restrictions and query timeouts.<br>• Developed large-dataset testing tools and updated execution traces for debugging. |
| **09/10/2026** | • Implemented a pre-synthesis evidence-validation stage to check requested subjects and facts, verified aliases, source provenance, and organization scope.<br>• Added distinct handling for partial answers, insufficient evidence, and retrieval failures, and corrected evidence ratings for unsupported answers.<br>• Completed regression testing with 311 passing tests and updated the project documentation.<br>• |
| **10/10/2026** | |

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| | **Holiday (Second Saturday)** |
| **11/10/2026** | **Holiday (Sunday)** |

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| **Next Week Plan** | • Test the updated system with real models and large datasets for answer accuracy, response time, and organization-only responses. |

**REFERENCE:**


https://microsoft.github.io/graphrag/get_started/

https://microsoft.github.io/graphrag/index/default_dataflow/

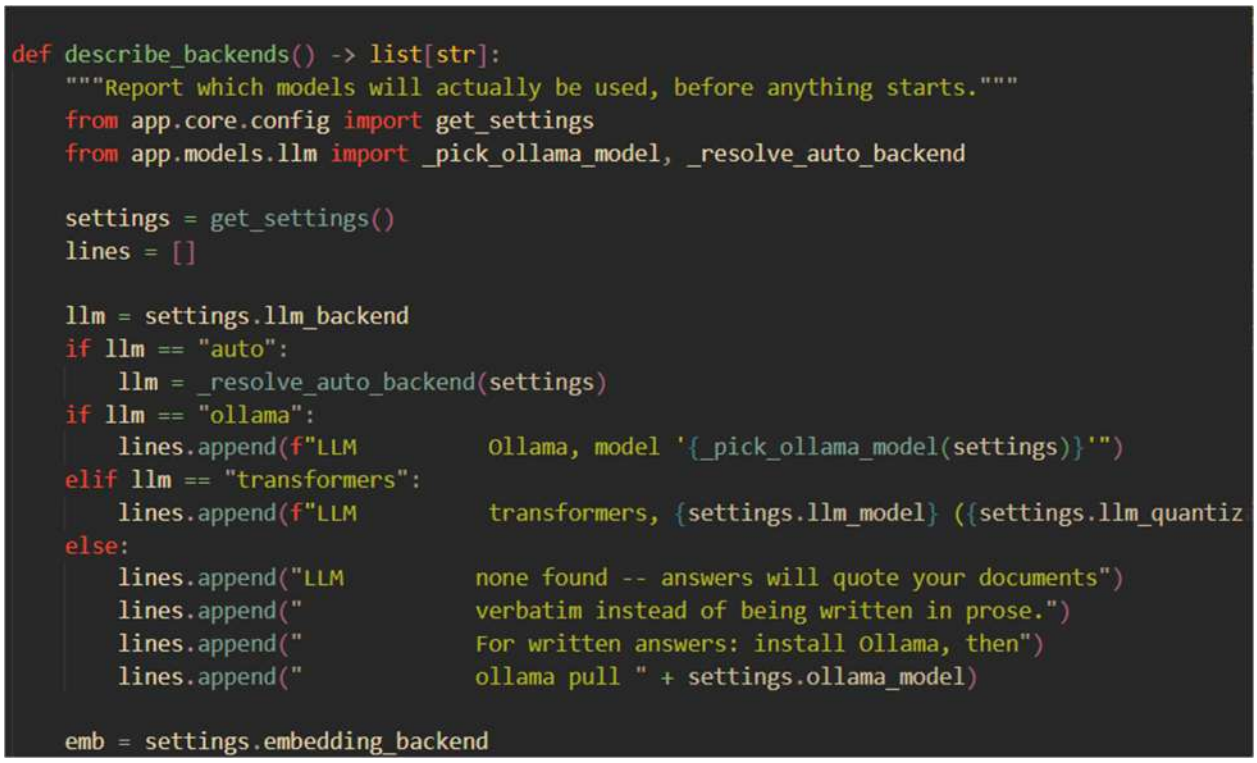
https://docs.ollama.com/

https://www.sqlite.org/docs.html

https://github.com/meta-llama/llama3

https://docs.python.org/3/library/sqlite3.html


**SCREEN SHOTS:**

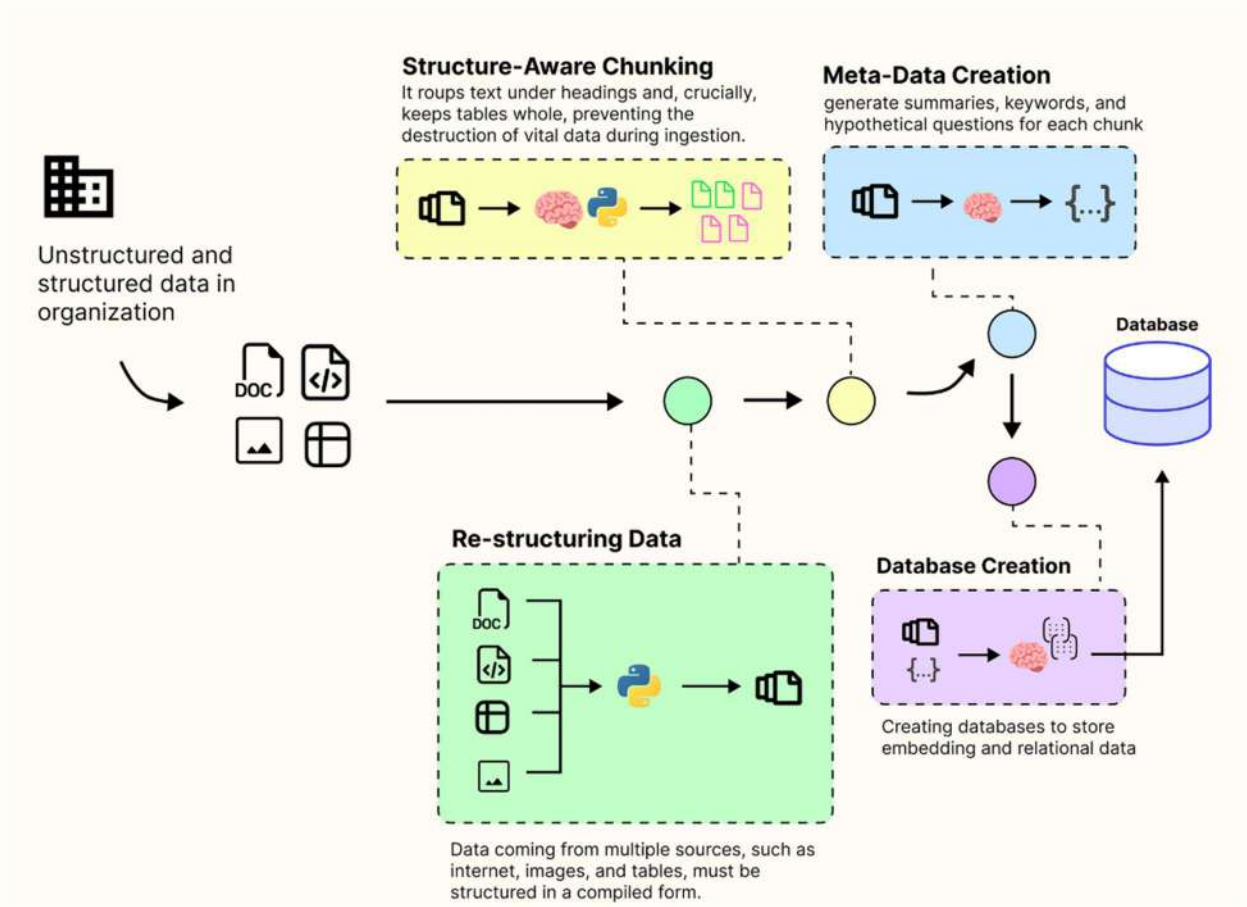


```
def describe_backends() -> list[str]:
    """Report which models will actually be used, before anything starts."""
    from app.core.config import get_settings
    from app.models.llm import _pick_ollama_model, _resolve_auto_backend

    settings = get_settings()
    lines = []

    llm = settings.llm_backend
    if llm == "auto":
        llm = _resolve_auto_backend(settings)
    if llm == "ollama":
        lines.append(f"LLM          Ollama, model '{_pick_ollama_model(settings)}'")
    elif llm == "transformers":
        lines.append(f"LLM          transformers, {settings.llm_model} ({settings.llm_quantiz
    else:
        lines.append("LLM          none found -- answers will quote your documents")
        lines.append("             verbatim instead of being written in prose.")
        lines.append("             For written answers: install Ollama, then")
        lines.append("             ollama pull " + settings.ollama_model)

    emb = settings.embedding_backend
```


Structure-Aware Chunking
It roups text under headings and, crucially, keeps tables whole, preventing the destruction of vital data during ingestion.
Meta-Data Creation
generate summaries, keywords, and hypothetical questions for each chunk
Unstructured and structured data in organization
DOC
Database
Re-structuring Data
DOC
Database Creation
Creating databases to store embedding and relational data
Data coming from multiple sources, such as internet, images, and tables, must be structured in a compiled form.